

An Investigation of Machine Learning Methods for Detecting Explainable Fake News

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Abstract—Higher amounts of digital spaces mean higher rates of spreading false information. Since the emergence of social media, their role has been altered, with some starting to spread incorrect news apart from other tasks. Given that there is such a high amount of daily information, it becomes hard to verify all the sources manually. The proposed model relies on machine learning techniques to recognize fake news and its reasoning behind that. Data will be taken from Kaggle and will consist of both categories of news – true and false. In the first stage, text cleaning is conducted, which involves removing noise, tokenizing, deleting stop words, and transforming it into numeric form using TF-IDF. Suddenly, three classic models emerge – Logistic Regression, Multinomial Naive Bayes, and Linear SVM, which receive their chance to train and evaluate themselves. Accuracy, as is known, plays a major role, yet equally important is the reasoning behind decisions, which is done via the same technique. In Logistic Regression, SHAP shows the importance of certain words, while the winner turns out to be Linear SVM, which wins by a mile.

Nonetheless, when considering this situation, the most important factor will be the effectiveness of the classifier. In turn, this means that the model can classify these two categories, ensuring the highest level of accuracy and making the fewest errors. With regard to gaining some insight into the problem and discovering the keywords associated with classifying that are most significant for this task, we need to look at the problem closer. It is interesting that, according to the analysis, the algorithms mentioned above can help detect fake news. Firstly, this way enables defining certain criteria for making decisions.

I. INTRODUCTION

With the increased use of digital media, how individuals access and utilize information has changed significantly. Online news and social media are becoming the primary sources of information of many people. This has led to quick access to information since it also allows quick spread of misguided information.

Basically, fake news is untrue or even false news presented as a legitimate news. Such content can potentially influence the mood of the masses, influence major decisions, and undermine trust in reliable information. Considering the huge amount of data that is generated everyday, it is impossible to be able to check all the news articles manually.

As a result, fake news detection algorithms Fake news detection systems that are based on ML and NLP have become a major source of research. An analysis of text, word selections and writing style can be conducted using machine learning models to identify whether a news article is authentic or a lie.

However, many machine learning systems are black boxes in that they do not provide explanations about their predictions. Explainable AI (XAI) aims to resolve this problem by providing a description of the way ML models arrive at their decisions.

II. OVERVIEW OF PRIOR WORK

One way to spot fake news involves older machine learning methods, though newer deep learning techniques also play a role. Starting with simpler tools, researchers rely on algorithms like Logistic Regression - clear, tested, built for sorting words at scale. Naive Bayes appears frequently, thanks to how it manages large sets of features without slowing down. Support Vector Machines step in when boundaries between true and false need sharper definition. The continued use of such models is because of their high performance and not because of the fact that they are sophisticated; they do exceptionally well under any circumstances.

The deep learning models make it possible to apply techniques such as the Convolutional Neural Networks for detecting the fake news. Moreover, using Recurrent Neural Networks can assist in recognizing patterns in the textual data. What makes this model effective is the fact that it ensures that the arrangement of the word sequence is taken into account. What is even better about such a technique is the Long Short-Term Memory.

But then again, despite the ease of operation in performing these operations, large volumes of data should be present, along with strong hardware capacity. However, it is difficult to comprehend the background behind such decisions as they rely on an elaborate algorithm.

There are many alternatives that can be pursued to solve the problem, such as, for example, SHAP, LIME, etc., using the methods of Explainable AI. This is necessary in order to comprehend the influence of individual variables.

III. DATASET DESCRIPTION

This experiment was based on the dataset obtained on Kaggle, which was a collection of categorized articles of fake and real news. The sample of the fabricated news articles contains about 23,481 items, and the sample of the factual news articles contains about 21,417 items.

Each news item also has such information as the title, main text, the theme, and the date. In this research the article text was the primary classification feature.

Before the models were trained, the two datasets were combined and annotated to tell the difference between fake and real news. Afterward, the united dataset was divided

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into training and testing datasets to efficiently determine the performance of the model.

IV. DATA PREPROCESSING

Raw text data often contain redundant items such as punctuations, web addresses, digits and special characters. These factors have a minimal impact on the classification process and should be eliminated.

Preprocessing of the text data was done as follows:

All text to be converted to lowercase. Removal of punctuation marks and other special symbols. Removing URLs and unnecessary characters. Stopwords removal – common words without any meaning. Tokenization – splitting of the text into individual words.

This preprocessing step will help to clean the data so that the model can focus only on important features.

V. EXTRACTION OF RELEVANT FEATURES OF TF-IDF

It is mostly done by the means of TF-IDF, which is a method in which a word is translated to a figure according to its importance in a particular document compared to the others. This algorithm does not concentrate on counting the appearances of a particular word; instead, it considers its occurrence in the collection of documents.

Usually, a term appears once in a document - we name that tally $TF(t, d)$. Every time a word shows up in different files, we mark its reach with $DF(t)$. The count of all those files goes into N . If something appears often in one spot but hardly anywhere else, it stands out more. Adjusting for that makes sorting things way more precise.

$$TFIDF(t, d) = TF(t, d) \times \log \left(\frac{N}{DF(t)} \right) \quad (1)$$

Word counts in one file matter most when they stand out across many files. Though common terms get less weight, rare ones gain importance through contrast. It all depends on where the term stands to measure its importance - more appearances, less places mean greater importance. This way, sorting will be made easier by differentiating. It is not about giving equal importance to every appearance; it is all about distribution.

VI. ML MODELS

A. Logistic Regression

Another method which can be employed to resolve binary classification issues through machine learning is known as Logistic Regression. This method utilizes a sigmoid function to ascertain the likelihood of an object belonging to a particular class.

$$\hat{y} = \frac{1}{1 + e^{-(\theta^T z + c)}} \quad (2)$$

B. Multinomial Naive Bayes

Naive Bayes Algorithm depends on the concept of probability. It employs the principles of Bayes theorem to enable it to make decisions. Since the attributes operate independently, there is an assumption that there is no correlation among them.

$$P(k | x) \propto P(k) \prod_{j=1}^m P(t_j | k) \quad (3)$$

C. Support Vector Machine

When it comes to the distinction of fake news from real news, it is at this point that the use of SVMs becomes relevant, particularly in the context of establishing the best boundaries. When there are many texts, SVMs linear classification proves relatively effective.

VII. EXPERIMENTAL SETUP

Splitting the data helped check how well the models worked. One part taught the system what to expect. While that happened, another piece stayed separate. Later, it checked if predictions held up on fresh examples. Learning came first, then real checking began

Getting text ready for machine learning meant turning it into numbers using TF-IDF vectorization. This method focused on the top 5000 frequent words only. Instead of keeping everything, less common terms were left out. The transformation allowed models to process language more effectively. Still, some nuance slipped through because rare words didn't make the cut.

VIII. EXPERIMENTAL RESULTS

Here's how each model did when spotting false stories. One after another, Logistic Regression stepped up first, then came Multinomial Naive Bayes, followed by the Linear Support Vector Machine. Each learned from data before being tested. Performance checks happened one at a time

Splitting the data came first - training versus test sets - before turning words into numbers through TF-IDF. Metrics like accuracy showed how it did, while precision, recall, and F1-score added detail; confusion matrices brought clarity at the end.

A. Model Accuracy Comparison

TABLE I
MODEL ACCURACY COMPARISON

Model	Accuracy
LR	98.65%
NB	92.87%
Linear SVM	99.51%

One thing stands out - the Support Vector Machine edged ahead when spotting false stories. Each model handled the task well, though accuracy wasn't equal across the board.

The following is the classification report of the Logistic regression model:

TABLE II
PERFORMANCE METRICES OF LOGISTIC REGRESSION MODEL

Class	recall	precision	f1-score
Unreliable	0.99	0.99	0.99
Reliable	0.99	0.98	0.99

TABLE III
CONFUSION MATRIX LOGISTIC REGRESSION.

	Classified as Fake	Classified as Real
Unreliable	4717	20
Reliable	24	4219

Table 1 below provides a confusion matrix of the Logistic Regression model.

Few mistakes popped up when the Logistic Regression model sorted through fake and real news, as shown by the confusion matrix. Most articles landed where they should.

TABLE IV
PERFORMANCE METRICS OF NAIVE BAYES MODEL

label	precision	recall	f1-Measure
Unreliable News	0.93	0.94	0.93
Reliable News	0.93	0.92	0.92

Whole Acc: 0.93 (according to 8980 samples)

TABLE V
CONFUSION MATRIX OF NAIVE BAYES.

	Classified as Fake	Classified as Real
Unreliable	4439	298
Reliable	342	3901

The reason why the Naive Bayes classifier made many mistakes than other models was that it assumed that there was independence among the features.

TABLE VI
PERFORMANCE EVALUATION OF SUPPORT VECTOR MACHINE

label	precision e	recall	f1 Measure
Unreliable	0.99	1.00	1.00
Reliable	1.00	0.99	0.99

Whole Acc: 1.00 according to 8980 test samples

TABLE VII
PERFORMANCE METRICES FOR SUPPORT VECTOR MACHINE

	Classified as Fake	Classified as Real
Unreliable	4717	20
Reliable	24	4219

For the model to perform efficiently, one must analyze the accuracy rate of precision, recall, and the F1 score. The precision gives information on the ratio of positively predicted outcomes. Unlike precision, recall indicates the total number of positively predicted outcomes. F1-score represents the result after multiplying the harmonic mean of precision and recall. If some classes have less data, macro

averaging guarantees that all the classes will be given equal weightage when calculating them.

The highest accuracy and the minimal number of errors in classification indicated that the SVM classifier was effective in distinguishing between fake and genuine news articles.

IX. EXPLAINABILITY ANALYSIS

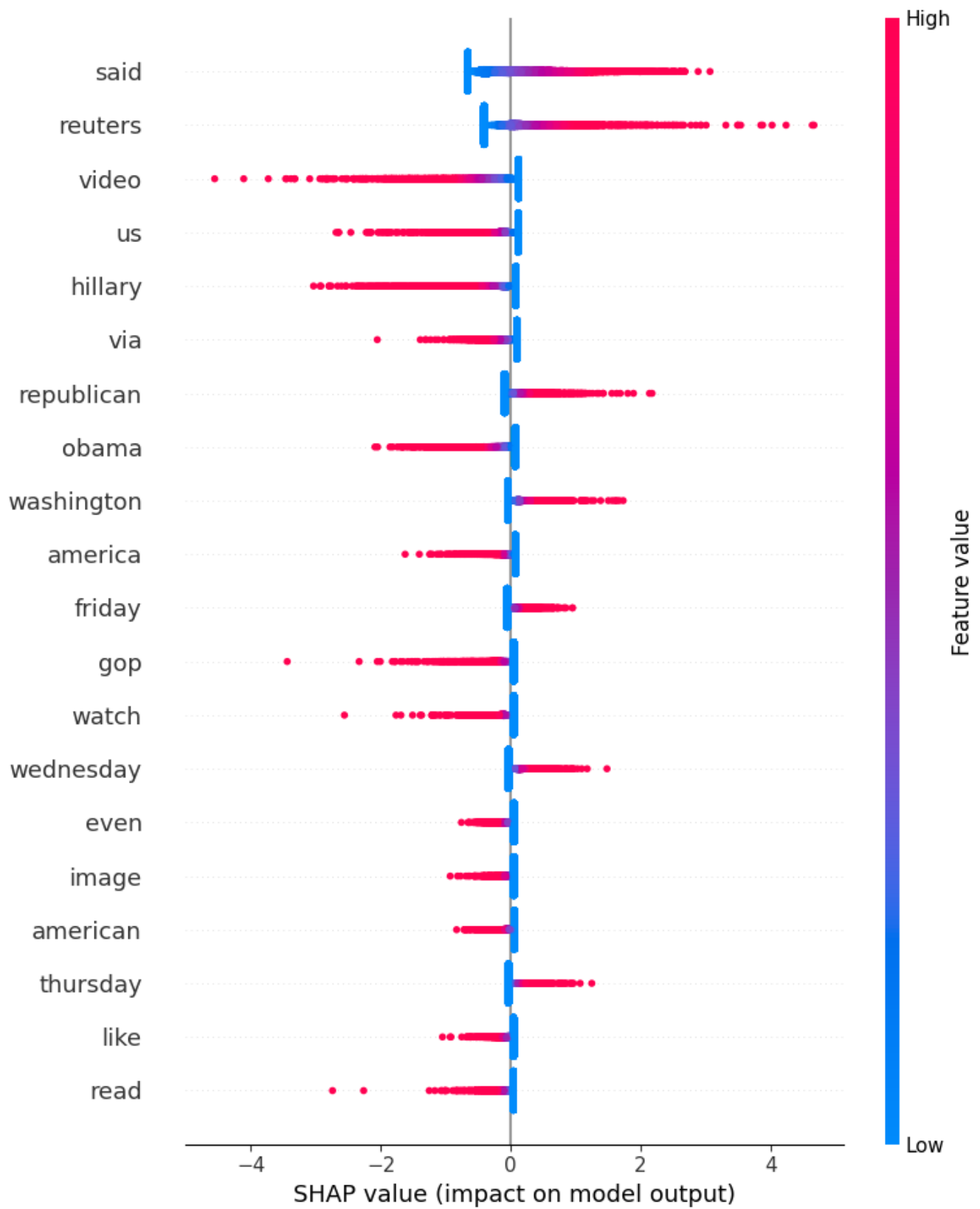
The reasoning behind the predictions of a model is essential to comprehend, and not just the predictions, especially in fields like detecting misinformation. Instead of considering the model an opaque system, which is explainable, the techniques can be explained to understand which factors influence the final decision. This enhances transparency and trust in the system.

This paper used SHAP (SHapley Additive exPlanations) to analyze the predictions made by the Logistic Regression model. SHAP is a technique of explainability, which is not dependent on the particular model used, that has its roots in cooperative game theory. It gives a weight to every feature which tells about its influence on the final prediction of the model. The computation of SHAP values will enable us to estimate which words in a news story have an impact on a prediction to make it fake or real. The SHAP summary plot is a general idea of the behavior of the model by depicting the most influential features across the entire dataset. The features with higher SHAP values have a stronger effect on model predictions. In detecting fake news, specific words are likely to appear more frequently in misleading, or sensational articles. Incorporating attention-seeking words, such as, video, breaking, watch and shocking, are often used in fake news headlines. These words, therefore, have a positive implication on forecasts of fake news.

On the other hand, authoritative news stories should provide references to official sources or follow common journalistic protocols. Words such as "Reuters," "said," "report," and "Washington" are usually present in the credible news articles, which facilitates the valid news predictions. As we look at these SHAP values, we can determine the linguistic aspects that characterize fabricated and authentic news.

Another advantage of SHAP is that it can provide both general and detailed explanations. Full-dataset explanations give an overview of the overall performance of the model, and the instance-specific explanations explain why a particular article was classified in this way. To give an example, when a news report employs a lot of sensational language and does not provide references to credible sources, the SHAP explanation will highlight these factors as contributing to the inaccurate forecast.

Naive Bayes and Support Vector Machine models can be explained in a number of ways. Naive Bayes prediction is performed based on the likelihood of words in various categories. Articles would be identified as fake more often in case they have words that are more common in fake articles than in real articles. The examination of these probabilities enables the researcher to identify the words that most affect classification choices.



SHAP summary plot shows the most influential characteristics in the ability to differentiate fake and real news, using a Logistic Regression model.

Similarly, the Support Vector Machine model involves the use of feature weights to determine the hyperplane which separates fake news and real news. The characteristics with higher positive correlations are useful in classifying items as true news, and the ones with negative correlations move the prediction towards finding them as fake. The analysis of these feature weights provides insight into how the SVM model distinguishes between the different categories of news articles.

In general, it is clear that in the process of explaining models, the relationship between model interpretability and linguistic characteristics in the news articles becomes apparent. This will be helpful in improving the reliability of automated detection methods of fake news, in addition to being useful for academic purposes.

X. DISCUSSION

The results clearly demonstrate that the accuracy of the algorithms depends on the representation of the features.

In this research, it is evident that machine learning algorithms have the ability to detect fabricated news. Out of all the machine learning algorithms, Support Vector Machine proved to be exceptional since it yielded an accuracy rate of 100 percent. The effectiveness of the algorithm implies that it efficiently processes high-dimensional text data.

Logistic Regression also demonstrated the impressive results with high precision and recall rates. One way it worked well was telling real from made-up stories. Another thing - it kept wrong guesses to a minimum as well as maintaining the misclassification rate. In addition, the Logistic Regression model allowed applying SHAP-based explainability techniques that provide valuable information concerning the factors influencing model predictions.

Compared to the other models, a bit less accurate turned out to be the Naive Bayes classifier. Feature independence assumed by the algorithm often leads to such results. With natural language data however, words are often used in related contexts to challenge this assumption and could reduce the accuracy of classification.

Confusion matrices provide more insight into the ways in which each model classifies data. Most of the predictions were correctly made and only a few predictions were not. Mistaken identifications were frequent due to the fact that fake news stories were disguised in the manner of original news stories or because original news stories were written in an exaggerated manner that was similar to those found in fake news.

Explainability analysis indicated that particular keywords can affect the predictions of the model significantly. The abundance of the words that are frequently used in the sensational headlines, such as the word video or breaking, is likely to tell that something is a fake news. On the other hand, the words related to trustworthy news reporting such as Reuters or said are used to anticipate true news. It is important to note that the results show the significance of language usage in distinguishing between genuine news stories and fake ones.

Generally, the results indicate that the classical approaches of machine learning remain highly effective and reliable for detecting fake news. Using approaches such as SHAP, these algorithms do not only provide accurate outputs but also explain their reasoning process.

XI. LIMITATIONS

But even so, even though the model is efficient enough in identifying cases of misinformation in an article, there exist a couple of challenges that should be identified with respect to the model. To begin with, the data used in coming up with the model is based on a particular source. As such, there will be some level of bias associated with the source, subject, or content of the article analyzed.

Secondly, the TF-IDF only looks at one-word tokens. Even though TF-IDF can be used to classify standard texts well, it fails to understand the full semantic context or meaning of complex texts. Consequently, such models may not be able to detect subtle language elements such as sarcasm, irony, or deceptive utterances because it involves a more thorough understanding of the situation.

Another limitation is that the proposed system relies solely on text to classify it. It is possible for misinformation to be conveyed through several methods, one of which is the use of photographs, films, and personal contact. These media are not considered in the model; therefore, the likelihood that the model will fail to detect misinformation is very high.

Finally, in contrast to interpreting the process behind the logistic regression model through the utilization of the SHAP values, no other machine learning algorithm utilized this approach for gaining an insight into its functioning. Although the probability of the value of the variable and its weight can offer some level of explanation, there are other approaches as well.

The limitations will act as evidence of the need to conduct additional studies that prove that the experiment conducted was indeed able to distinguish fake news from genuine news.

XII. FUTURE WORK

As far as the future study in identifying the fake news is concerned, the researchers can explore the use of various kinds of neural networks. There is no need for them to be constrained by the conventional forms of neural network configurations only; instead, what they may perhaps think of doing is exploring the use of more advanced neural network configurations such as the one which is known as the convolutional neural network, which has the potential to detect patterns in the structure of the data. Besides this, yet another neural network configuration which may serve their purpose well is the recurrent neural network.

One other potential field of research that may be considered involves the integration of multiple types of data. Fake news in real-life scenarios typically contains multiple images, videos, and social media data. The integration of textual as well as visual and contextual data will help enhance the accuracy of fake news detection algorithms.

Further research could investigate using bigger and more varied datasets to enhance the models' ability to generalize. It is possible to reduce bias and enhance the robustness of models by using training datasets of different news sources and languages.

Moreover, other studies might focus on improving the transparency of models. Even though the interpretation of the Logistic Regression model in this study was done using SHAP values, more elaborate explainability algorithms may improve the interpretation of deep learning models and provide deeper explanations of the classification results.

Application of fake news detection systems in real-time settings such as social media monitoring systems could be more effective in identifying and curbing the spread of fake information.

XIII. CONCLUSION

In this paper, I present a machine learning model that describes its process of detecting fake news, through conventional text classification techniques. With the rapid development of online news outlets and social media, the proliferation of fake news has become an important global issue. Therefore, computerized detection tools are essential in identifying fake materials and reducing the spread of misinformation.

From a collection of tagged false and true news pieces, one trial shaped how three learning systems performed - Logistic Regression, Multinomial Naive Bayes, alongside Linear Support Vector Machine. To feed written content into these frameworks, techniques like removing frequent filler words, breaking down sentences into fragments, along with term weighting through frequency-inverse document scoring turned language into numbers. These moves let each model spot distinctive word behaviors within the articles they examined.

Surprisingly, older types of algorithms handled the task of spotting false news quite effectively. Outperforming the rest, the Support Vector Machine stood out during testing. Close behind came Logistic Regression - nearly on par despite simpler design. Independence assumptions did slow down Naive Bayes a bit. Still, it managed to deliver workable outcomes overall.

A key aspect of this work is employing explainability methods to gain insight into model predictions. SHAP (SHapley Additive exPlanations) was applied to find out the most significant characteristics. The result of this analysis indicates that the language frequently used for sensational headlines is normally linked with the prediction of fake news, while the terminologies used for credible sources are used to classify genuine news.

Overall, the proposed framework shows that conventional machine learning models, along with explainability techniques, can provide both accurate and interpretable results. Such systems can contribute to the increased trust in automated verification tools to a considerable extent and prevent fake information spreading.

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